

Spin^h structures on Symmetric Spaces

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1 Introduction

We denote by $\mathrm{SO}(n)$ the special orthogonal group of $n \times n$ matrices. The spin group $\mathrm{Spin}(n)$ is a double cover of $\mathrm{SO}(n)$. We construct generalized spin groups

$$\mathrm{Spin}^r(n) = (\mathrm{Spin}(n) \times \mathrm{Spin}(r))\mathbb{Z}_2,$$

where we view $\mathbb{Z}_2$ as the subgroup of $\mathrm{Spin}(n) \times \mathrm{Spin}(r)$ generated by $(-1, -1)$. When $r = 1$, this is the original spin group $\mathrm{Spin}(n)$. When $r = 2$, these are the complex spin groups, denoted $\mathrm{Spin}^c(n)$. We are particularly interested in the $r = 3$ case, which are the quaternionic spin groups $\mathrm{Spin}^h(n) = (\mathrm{Spin}(n) \times \mathrm{Sp}(1))\mathbb{Z}_2$.

For all r , there is a double cover

$$\varphi^{r,n} : \mathrm{Spin}^r(n) = (\mathrm{Spin}(n) \times \mathrm{Spin}(r))/\mathbb{Z}_2 \rightarrow \mathrm{SO}(n) \times \mathrm{SO}(r),$$

which is given by the morphisms $\mathrm{Spin}(n) \rightarrow \mathrm{SO}(n)$ and $\mathrm{Sp}(1) = \mathrm{Spin}(r) \rightarrow \mathrm{SO}(r)$ (see [AL23]).

Recall that a spin structure on a principal $\mathrm{SO}(n)$ -bundle P is a principle $\mathrm{Spin}(n)$ -bundle Q with a double cover $Q \rightarrow P$ which is equivariant with respect to $\mathrm{Spin}(n) \rightarrow \mathrm{SO}(n)$. Analogously, a spin^h structure on P is a principal $\mathrm{Spin}^h(n)$ -bundle Q and a principal $\mathrm{SO}(3)$ -bundle E , along with a double cover $Q \rightarrow P \times E$ which is equivariant with respect to $\varphi^{3,n} : \mathrm{Spin}^h(n) \rightarrow \mathrm{SO}(n) \times \mathrm{SO}(3)$. We refer to [Fri00] for discussion on spin structures and [AM21] for further discussion of spin^h structures. For an oriented Riemannian manifold M , we denote by F_M the principal $\mathrm{SO}(n)$ -bundle of oriented orthonormal frames on M . We say that M has a spin^h structure if F_M does.

Let G be a Lie group and $H \subset G$ a Lie subgroup. The homogeneous space G/H carries a transitive action of G , with isotropy subgroup H . On such manifolds, we are interested in the existence of spin and spin^h structures which are invariant under the action of G . If G/H is of dimension d , we denote by σ the isotropy representation $\sigma : H \rightarrow \mathrm{SO}(d)$ induced by the differential of the map $\mu_h : G/H \rightarrow G/H$ for any $h \in H$.

By [SKL22][Proposition 1.3], the existence of a G -equivariant spin structure on G/H is equivalent to a lift $\tilde{\sigma}$ of the isotropy representation σ to $\mathrm{Spin}(d)$. Since $\mathrm{Spin}(d)$ is a covering space of $\mathrm{SO}(d)$, this occurs exactly when the induced morphism $\sigma_{\#} : \pi_1(H) \rightarrow \pi_1(\mathrm{SO}(d))$ is trivial.

A similar method can be used to determine the existence of G -equivariant spin^h structures. Applying [AL23][Theorem 3.10] to the $r = 3$ case, a G -invariant spin^h structure on G/H exists if and only if there exists a Lie group homomorphism $\varphi : H \rightarrow \mathrm{SO}(3)$ such that $\sigma \times \varphi$ lifts along $\varphi^{3,n}$:

$$\begin{array}{ccc} H & \xrightarrow{\sigma \times \varphi} & \mathrm{SO}(n) \times \mathrm{SO}(3) \\ & \searrow \varphi^{3,n} & \nearrow \\ & \mathrm{Spin}^h(n) & \end{array}$$

Since $\mathrm{Spin}^h(n)$ is a double cover of $\mathrm{SO}(n) \times \mathrm{SO}(3)$, this occurs if and only if

$$(\sigma \times \varphi)_{\#}(\pi_1(H)) \subseteq \varphi_{\#}^{3,n}(\pi_1(\mathrm{Spin}^h(n))).$$

We are interested in classifying homogeneous spaces which do or do not have spin^h structures. In particular, we look at Riemannian symmetric spaces, which are homogeneous spaces G/H for which there exists an automorphism $\rho : G \rightarrow G$ which is an involution and satisfies

$$(G^{\rho})^0 \subset H \subset G^{\rho},$$

where G^ρ denotes the set of fixed points and $(G^\rho)^0$ is the connected component of its identity. Not all symmetric spaces are spin, and it is also possible that a spin symmetric space admits no G -invariant spin structure. We will see in the following sections that there are spaces with a G -invariant spin^h structure but no spin structure.

Some trivial examples are as follows:

Example 1.1. If H is simply connected, the isotropy representation σ necessarily induces a trivial map on fundamental groups. Thus in this case G/H always has a G -invariant spin structure.

Example 1.2. Suppose that G/H has a G -invariant spin structure, i.e. that the map $\sigma_\#$ of fundamental groups is trivial. Then we may take $\varphi : H \rightarrow \text{SO}(3)$ to be the trivial homomorphism, so that the image of $\pi_1(H)$ under $(\sigma \times \varphi)_\#$ is trivial. This is always in the image of $\pi_1(\text{Spin}^h(n))$ under $\varphi_\#^{3,n}$, so there is a G -invariant spin^h structure. Under the correspondence described in [AL23][Theorem 3.10], this yields the spin^h structure on G/H induced by the given spin structure.

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2 Grassmannians

Two examples of homogeneous spaces are the real and complex Grassmannians. It is known that in the real case $\text{Gr}_{\mathbb{R}}(k, n+k)$, a spin structure exists if and only if $n+k$ is even, and it is G -invariant if and only if both k and n are even. In the complex case, a spin structure exists if and only if $n+k$ is even, and it is always G -invariant.

2.1 $\text{SO}(n+k)/\text{SO}(n) \times \text{SO}(k)$

In this case, we view the subgroup $H = \text{SO}(n) \times \text{SO}(k)$ as the subgroup of block diagonal matrices in $G = \text{SO}(n+k)$ with the two blocks being elements of $\text{SO}(n)$ and $\text{SO}(k)$, respectively. This space is isomorphic to the real Grassmannian, which parametrizes all k -dimensional vector subspaces of $\mathbb{R}^{n+k}$. Since the dimension of $\text{SO}(m)$ is $\frac{m(m-1)}{2}$, we compute that $\dim G/H = nk$. The fundamental group $\pi_1(H)$ is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, generated by classes $[\gamma_1(t)]$ and $[\gamma_2(t)]$. If the isotropy representation σ lifts to $\text{Spin}^h(nk)$, there is a diagram

$$\begin{array}{ccc}
 & \text{Spin}^h(nk) & \\
 \tilde{\sigma} \nearrow & & \searrow \varphi_{3,nk} \\
 H = \text{SO}(n) \times \text{SO}(k) & & \text{SO}(nk) \times \text{SO}(3) \\
 \sigma \searrow & & \swarrow \text{pr}_1 \\
 & \text{SO}(nk) &
 \end{array}$$

We project $\text{SO}(nk) \times \text{SO}(3) \rightarrow \text{SO}(3)$ to obtain a map $H \rightarrow \text{SO}(3)$. This induces a Lie algebra homomorphism $\mathfrak{so}(n) \times \mathfrak{so}(k) \rightarrow \mathfrak{so}(3)$.

First considering the case where both n and k are greater than 4, both $\mathfrak{so}(n)$ and $\mathfrak{so}(k)$ are simple. Then the only proper ideals of $\mathfrak{so}(n) \times \mathfrak{so}(k)$ are $\mathfrak{so}(n)$ and $\mathfrak{so}(k)$, so dimension constraints imply that the kernel of this homomorphism must be all of $\mathfrak{so}(n) \times \mathfrak{so}(k)$. Now $\text{SO}(n) \times \text{SO}(k)$ is connected and thus generated by the exponential map, so $H \rightarrow \text{SO}(3)$ must be trivial. From [AL23][Proposition 2.1], the image of $\pi_1(\text{Spin}^h(nk))$ under $\varphi_\#^{3,nk}$ is the subgroup $\langle (1,1) \rangle \subset \mathbb{Z}_2 \times \mathbb{Z}_2$. Thus the generators $[\gamma_1(t)], [\gamma_2(t)]$ of $\pi_1(H)$ must be sent to 0 in $\pi_1(\text{SO}(nk) \times \text{SO}(3))$. But their images under $\sigma_\#$ are $n \bmod 2$ and $k \bmod 2$, respectively, so this occurs exactly when n, k are both even. In this case G/H also has a G -invariant spin structure.

We note that for $k = 1$, this is the case of $\text{SO}(n+1)$ -invariant spin structures on the sphere S^n , which we know to exist. Thus it remains to consider $k = 2, 3, 4$.

When $k = 3$, we wish to find a Lie group homomorphism

$$\varphi : SO(n) \times SO(3) \rightarrow SO(3)$$

such that $\sigma \times \varphi$ lifts to $\text{Spin}^h(3n)$. By the lifting property of covering spaces, such a lift exists if and only if

$$(\sigma \times \varphi)_\#(\pi_1(SO(n) \times SO(3))) \subset \varphi_\#^{3,3n}(\pi_1(\text{Spin}^h(3n))) = \langle (1, 1) \rangle.$$

Recall that the generators of $\pi_1(H)$ are given by

$$\gamma_1(t) = (\alpha_n(t), \text{Id}_3), \gamma_2(t) = (\text{Id}_n, \alpha_3(t)).$$

Their images under $\sigma_\#$ are 1 and $n \bmod 2$, respectively. We wish for their images under $\varphi_\#$ to be the same. Assuming that n is not equal to 2 or 4, passing to the level of Lie algebras, we conclude in the same way as above that φ must be trivial on at least one of the factors $SO(n)$ and $SO(3)$. If φ depends solely on the factor of $SO(3)$, it sends the generator $[\gamma_1(t)]$ to 0, so it cannot lift. If φ depends solely on $SO(n)$, we recall that there are no nontrivial Lie algebra homomorphisms $SO(n) \rightarrow SO(3)$ for $n \geq 5$. Furthermore, since φ will send $[\gamma_2(t)]$ to 0, so n must be even. Thus in this case there is no value of n for which $\sigma \times \varphi$ lifts.

Now we show that for $k = 2$, a G -invariant Spin^h structure exists. We only need to consider odd values of n . Using the same generators of H as before, under $\sigma_\#$, $[\gamma_1(t)]$ and $[\gamma_2(t)]$ are sent to 0 and 1, respectively. We take the map $\varphi : SO(n) \times SO(2) \rightarrow SO(3)$ induced by the natural inclusion $SO(2) \rightarrow SO(3)$. This inclusion sends the generator $[\alpha_2(t)]$ of $\pi_1(SO(2))$ to the generator of $\pi_1(SO(3))$. Then see that under $\varphi_\#$, $[\gamma_1(t)]$ is sent to 0 since it depends only on the factor of $SO(2)$. Similarly, it sends $[\gamma_2(t)]$ to 1. Thus $\sigma \times \varphi$ lifts to $\text{Spin}^h(nk)$.

Finally, a lift also exists for $k = 4$ and odd n . We again wish for a map φ which sends $[\gamma_1(t)]$ to 0 and $[\gamma_2(t)]$ to 1. Starting on the level of Lie algebras, there is a decomposition $\mathfrak{so}(4) = \mathfrak{so}(3) \times \mathfrak{so}(3)$. Taking the Lie algebra homomorphism $\mathfrak{so}(4) \rightarrow \mathfrak{so}(3)$ by projection onto the first factor, we integrate to obtain a map $SO(4) \rightarrow SO(3)$. This induces a nontrivial map of fundamental groups. Then we may define φ to be induced by this homomorphism, so that it satisfies the desired conditions.

2.2 $SU(n+k)/S(U(n) \times U(k))$

The subgroup $H = S(U(n) \times U(k))$ of $G = SU(n+k)$ consists of matrices of the form

$$X = \begin{pmatrix} U(n) & 0 \\ 0 & U(k) \end{pmatrix}$$

with $\det X = 1$. This space is isomorphic to the complex Grassmannian. We compute that $\dim G/H = 2nk$. Recall that the Lie algebra $\mathfrak{u}(n)$ of $U(n)$ consists of $n \times n$ skew-Hermitian matrices, and $\mathfrak{su}(n)$ consists of $n \times n$ skew-Hermitian matrices with trace 0. Furthermore, $\mathfrak{su}(n)$ is a simple Lie algebra for all $n \geq 2$. Then we may see that the Lie algebra $\mathfrak{h}$ of H consists of matrices of the form

$$\begin{pmatrix} X_n & 0 \\ 0 & X_k \end{pmatrix}$$

such that $X_n \in \mathfrak{u}(n)$, $X_k \in \mathfrak{u}(k)$, and $\text{Tr}(X_n) = -\text{Tr}(X_k)$. Then we can write

$$\mathfrak{h} = \mathfrak{su}(n) \times \mathfrak{su}(k) \times \mathbb{R},$$

where the factor of $\mathbb{R}$ controls the trace of X_n . Now in the same way as the previous case, from the Lie group homomorphism $H \rightarrow SO(3)$, we obtain a Lie algebra homomorphism $\psi : \mathfrak{h} \rightarrow \mathfrak{so}(3)$. The dimension of $\mathfrak{su}(n)$ is $n^2 - 1$, and $\dim \mathfrak{so}(3) = 3$. Hence the kernel of ψ must either be all of $\mathfrak{h}$, or (without loss of generality) k must equal 2 and $\ker \psi$ must be $\mathfrak{su}(n) \times \mathbb{R}$.

First consider the case where $\ker \psi = \mathfrak{h}$. H is connected, so $H \rightarrow SO(3)$ is trivial. The fundamental group $\pi_1(H)$ is isomorphic to $\mathbb{Z}$ and generated by the class of a loop $[\gamma]$ such that $\sigma_*([\gamma]) = n+k \bmod 2$. Assuming that $2nk \geq 3$, again by [AL23][Proposition 2.1],

$$\varphi_\#^{3,2nk}(\pi_1(\text{Spin}^h(2nk))) = \langle (1, 1) \rangle \subseteq \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Since $H \rightarrow \mathrm{SO}(3)$ is trivial, the image of the fundamental group must be $(0, 0)$, so $n + k$ is even. Thus this corresponds to the case where a spin structure exists of G/H and induces the spin^h structure.

Now suppose that $k = 2$ and $\ker(\psi) = \mathfrak{su}(n) \times \mathbb{R}$. It only remains to consider odd values of n . Then $\sigma_{\#}([\gamma]) = 1 \in \mathbb{Z}_2$. We wish to find

$$\varphi : H = \mathrm{S}(\mathrm{U}(n) \times \mathrm{U}(2)) \rightarrow \mathrm{SO}(3)$$

such that $\varphi_{\#}([\gamma]) = 1$. We recall that there is an isomorphism $PU(2) \simeq \mathrm{SO}(3)$, and the projection $U(2) \rightarrow (2) = U(2)/U(1)$ is nontrivial on the fundamental group of $U(2)$. Now we may take φ to be induced by this projection on the second factor of H and trivial on the first factor, and φ is also nontrivial on the fundamental group of H . Thus in this case G/H has a spin^h structure.

3 Symmetric spaces

It is known which irreducible symmetric spaces of type I are spin and also when the spin structures are G -invariant. We wish to know if any additional spaces of this form have a spin^h structure.

All almost-quaternionic manifolds are spin^h (see [AM21]). This motivates the question of whether the same can be said for Wolf spaces, or quaternion-Kähler symmetric spaces. Such spaces with positive Ricci curvature can be classified in terms of quotients of simple Lie groups. The two Grassmannian cases above are Wolf spaces, and the other classical case is that of $\mathrm{Sp}(n+1)/\mathrm{Sp}(n) \cdot \mathrm{Sp}(1)$.

3.1 $\mathrm{Sp}(n+1)/\mathrm{Sp}(n) \cdot \mathrm{Sp}(1)$

A related space is the sphere $S^{4n+3} \simeq \mathrm{Sp}(n+1) \cdot \mathrm{Sp}(1)/\mathrm{Sp}(n) \cdot \mathrm{Sp}(1)$. We have the exact sequence

$$\mathrm{Sp}(1) \rightarrow S^{4n+3} \rightarrow \mathbb{H}\mathbb{P}^n = \mathrm{Sp}(n+1)/\mathrm{Sp}(n) \times \mathrm{Sp}(1),$$

and in the sphere case, the inclusion $f_n : \mathrm{Sp}(n) \cdot \mathrm{Sp}(1) \hookrightarrow \mathrm{Sp}(n+1) \cdot \mathrm{Sp}(1)$ is given by

$$f_n([A, z]) = \left[\begin{pmatrix} z & 0 \\ 0 & A \end{pmatrix}, z \right].$$

We refer to [SKL22][Lemma 2.3] for the explicit details of this construction. Thus in our case, we may work with the block diagonal product of $\mathrm{Sp}(n)$ and $\mathrm{Sp}(1)$. We have the loop $\gamma(t) = [e^{\pi it} \mathrm{Id}_n, e^{\pi it}] \in H$, which yields a generator $[\gamma]$ of $\pi_1(H)$. For the same reasons as the sphere case, $\sigma_{\#}([\gamma]) = n+1 \bmod 2$. When n is odd, the isotropy representation lifts and there is a spin structure. When n is even, we construct $\varphi : \mathrm{Sp}(n) \cdot \mathrm{Sp}(1) \rightarrow \mathrm{SO}(3)$ sending $[\gamma]$ to $1 \in \mathbb{Z}_2$.

The complexified Lie algebra $\mathfrak{h}^{\mathbb{C}}$ is isomorphic to $\mathfrak{sp}_{\mathbb{C}}(2n) \times \mathfrak{sp}_{\mathbb{C}}(2)$. Recalling that $\mathfrak{sp}_{\mathbb{C}}(2) \simeq \mathfrak{so}(3)$, consider the Lie algebra homomorphism $\mathfrak{h}^{\mathbb{C}} \rightarrow \mathfrak{so}(3)$ induced by this projection. Now using the description of $\gamma(t)$ above, we see that the induced Lie group homomorphism is nontrivial on the fundamental group. Thus there is a spin^h structure.

3.2 $\mathrm{SO}(2n)/\mathrm{U}(n)$

This space always has a spin structure, but it is G -invariant only when n is odd. We show that when n is even, there is a spin^h structure. We view $H = U(n)$ inside $G = \mathrm{SO}(2n)$ by writing complex numbers $a + bi$ as real matrices

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

For even n , $\sigma_{\#}$ sends the generator of $\pi_1(H)$ to $1 \in \mathbb{Z}_2$. Thus we wish to find $\varphi : U(n) \rightarrow \mathrm{SO}(3)$ with $\varphi_{\#}(1) = 1$. Consider the following composition of morphisms:

$$U(n) \xrightarrow{\det} U(1) \hookrightarrow \mathrm{SO}(2) \hookrightarrow \mathrm{SO}(3).$$

The loop whose class generates $\pi_1(U(n))$ is $\gamma(t) = \mathrm{diag}(e^{\pi it}, 1, \dots, 1)$. The determinant map sends this loop to a generator for $\pi_1(U(1))$, and then we see that the composition $U(n) \rightarrow \mathrm{SO}(3)$ induces a nontrivial map on the fundamental groups. Thus there exists a spin^h structure.

3.3 Exceptional Lie Groups

With the exception of $F_4/\mathrm{Sp}(3) \cdot \mathrm{Sp}(1)$, the remaining Wolf spaces have G -invariant spin structures.

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